

Structuring a Development Process for AI/ML Projects: A Look into Industry Driven Issues

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Abstract: AI and ML projects differ in several aspects from traditional software development, so the traditional and agile development process cannot overcome the number of difficulties and aspects of these types of software development. In many cases, the AI and ML development cycles, even though representing just a fraction of the system, are made using non-standardized and inadequate development processes that cannot guarantee the developed product's quality and effectiveness. This paper intends to elucidate and shed light on this issue, showcasing a novel development lifecycle process for AI/ML contexts that accounts for the quality activities and results to improve the project results. This research is developed in collaboration with the industry and is the first contribution on the developed process grounded on empirical information and validated on an empirical quality process. We present the first version of the development lifecycle process for AI/ML enabled systems and also present a initial evaluation of the developed artifact from the view of industry specialists, presenting the elicitation of further research and development for the next iteration.

1 INTRODUCTION

The increasing pressure and desire among software development companies and industries to integrate AI (Artificial Intelligence) and ML (Machine Learning) systems into the core of their solutions are becoming pervasive in day-to-day operations (Martínez-Fernández et al., 2022). The rapid advancement of AI/ML technologies has made them integral to various domains, including finance, healthcare, and automation, further accelerating their adoption in software development. However, much of this research focuses on performance improvements without adequately considering how these systems are being developed (Gezici and Tarhan, 2022).

Despite the versatility of AI and ML systems in solving a variety of problems, their development process differs significantly from traditional software development practices, such as those used for websites, mobile applications, and games (Ozkaya, 2020). Un-

like traditional software, AI/ML projects rely heavily on data collection, pre-processing, and model validation, requiring additional steps before actual development begins. The unique nature of AI-based systems necessitates additional attributes, steps, and considerations that are not typically accounted for in conventional software development cycles (Ozkaya, 2020). Notably, quality aspects are included only in the last steps of the AI model development and integration/deploy phases, where quality analysts can analyze and verify the quality of the developed models to be integrated on the final version of the software.

This research is crucial because current methodologies do not fully accommodate these differences, leading to challenges in project estimation, planning, and execution. As AI and ML continue to evolve, addressing these gaps becomes essential for ensuring the success of software projects that integrate these technologies.

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2 PROBLEM DEFINITION

AI-based systems require extensive pre-work, validation, and data considerations that impact the overall measurement of project size and complexity (Giray, 2021). Unlike traditional software development, where the scope is relatively well-defined, AI/ML projects often face uncertainties related to data availability, model performance, and computational requirements — aspects frequently overlooked by clients, product owners, and project managers when estimating project scope and resources (Nascimento et al., 2020). The inability to accurately measure this work is evident in agile and traditional processes, which generally do not account for the necessary preparatory tasks before model development or the technical feasibility assessment of a project (Polkowski et al., 2019; Lorenzoni et al., 2021). This gap in estimation poses risks to project delivery, including underestimated development efforts, unexpected resource requirements, and budget miscalculations (Ozkaya, 2020). For example, AI projects often require significant effort in data preparation and model training, which may not be initially considered, leading to project delays and cost overruns. Furthermore, developers working on AI projects may require specialized skills in data science, machine learning engineering, and model optimization, which are often not planned for in conventional project estimations.

In addition, AI/ML development processes do not account for quality evaluation steps and the quality metrics evaluated in each step of model development. Traditional software development processes consider validation steps and quality evaluation activities for each version and include quality analysts on the software definition and development steps (Lwakatare et al., 2020). Given these challenges, this research aims to investigate the existing development patterns in the industry and propose a new software development framework tailored for AI and ML projects, considering quality steps and analysis to improve software integration. This framework will help identify crucial steps that are often invisible to clients and project estimators, ensuring better project planning and execution. By providing a structured method for assessing AI/ML project complexity, this study aims to bridge the gap between software engineering best practices and the unique requirements of AI-driven development.

3 RELATED WORK

3.1 Software Development Lifecycle for AI-Based Systems

Lorenzoni et al. (2021) (Lorenzoni et al., 2021) note that software developers face several challenges in ML systems development, as they often engage in ad-hoc practices that lead to problems with system delivery and requirements adherence. Although traditional processes could partially benefit ML projects, ML systems require different steps, data, and scopes. Martínez-Fernández et al. (2022) (Martínez-Fernández et al., 2022) further identified that, despite considerable research on AI-based systems, there is limited synthesized knowledge on Software Engineering (SE) approaches for building, operating, and maintaining them. A great part of the identified SE approaches are related to software testing and quality, while areas such as software maintenance lack studies. The authors also identified that data-related issues are the most prevalent, and a final SE process approach is needed for AI-based systems. Gezici and Tarhan (2022) (Gezici and Tarhan, 2022), through a Systematic Literature Review (SLR), found that much research has proposed ad-hoc solutions for Quality Analysis (QA) and Quality Control (QC) of AI-based software, yet a consolidated and structured process remains absent.

3.1.1 Traditional AI/ML Development Process

The current process for developing AI/ML projects in the industry comprises three main phases: **Research and Data Sampling**, **Algorithm Development**, and **Solution Deployment**. When exploring the phases contained inside them, as depicted in Figure 1, the only step where a quality evaluation process is performed is after the AI/ML model is complete and finished for deployment. This raises several issues considering the Quality Analysis process throughout model development, where quality is not analyzed and issues are identified only in late steps, producing higher-impact problems that consume more time, effort, and budget (Lwakatare et al., 2020). The main concern is therefore to develop and include quality verification and support steps that will identify and prevent issues earlier in the process, making AI/ML models more precise, with more quality, and more fit to the defined system requirements and features.

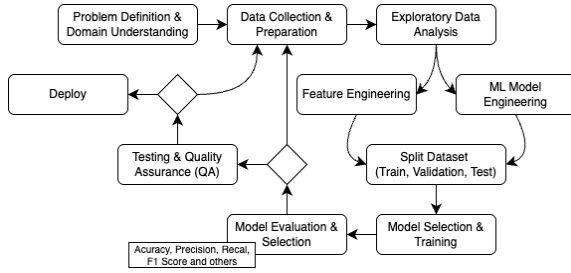


Figure 1: Traditional AI/ML model development process for AI/ML enabled software.

4 INDUSTRY GATHERED KNOWLEDGE

The comprehension of the problem definition proposed in this paper is the result of the knowledge gathered from AI/ML projects developed in the past decade in a R&D company, where one of the authors contributes as a Quality Analyst in AI/ML related projects. All the knowledge collected and the proposed emerging solution were developed in collaboration with company experts and partners who have participated in and developed AI/ML models throughout the past years.

4.1 Industry Interviews

We conducted interviews with four specialists from the company who have worked on AI/ML model development projects in the past 10 years. Project scope and expertise varied for each specialist. Interviews were recorded and transcribed for annotation and classification. All interview and data analysis followed the process defined by Cruzes & Dybå (2017) (Cruzes and Dybå, 2017). The content analysis process was conducted upon the transcribed interviews; audio files were used for discussions when transcription misunderstandings were identified. To control knowledge and classification bias, a random sample of transcribed interview sections were selected and a cross-validation was performed by two authors. All data was evaluated and managed under the Informed Consent term signed by all interviewed collaborators.

4.1.1 Specialists Characterization

- **Specialist 1 (S1):** Software Manager in NLP and AI projects, with more than 15 years of experience on development and coordination of NLP and AI projects in the company;
- **Specialist 2 (S2):** AI/ML Project Tech Leader at the R&D company, PhD candidate in AI re-

search, main developer and coordinator in AI/ML projects, with more than 7 years of experience on AI/ML software projects;

- **Specialist 3 (S3):** QA team manager of the business and development unit, head of the QA team in AI/ML and NLP projects, with more than 5 years of experience in test procedures and quality patterns;
- **Specialist 4 (S4):** CEO and Site Manager of the R&D institute, focused on developing and delivering software systems integrated with AI and ML models of varied scopes, from visual modeling, line prediction, LLM development, among others. S4's objective is to improve the quality of delivered systems while establishing a reference industry for AI system development.

4.1.2 Overall Interview Results

Specialist S1 specified that the main issue faced in the project development cycle was that: "...after model development QA team validates system integration ... and raised issues often relate to the AI model ..." - S1. S1 also raised the concern that there were no known solutions that addressed this gap in the development process.

Specialist S2 stated that the QA process is one of the most important software development steps; however, QA teams were not present during AI model definition and development: "...even though, I think that QA Teams need to develop processes that can support the AI model definition and development..." - S2. S2 identified that the most valuable QA steps would include data quality reviews and data analysis, reducing bias and making data sectioning more robust and larger overall.

Specialist S3 stated that QA team members missed the necessary base knowledge on AI development, preventing them from supporting the development team and model validation process. S3 also noted that no AI model development process was in place to guide validation and verification steps on the developed AI model and general outcomes. Furthermore, S3 agreed that a software process that contains quality analysis and verification steps designed for it alone could greatly impact the development of the needed knowledge overall on the team. Moreover, S3 believes that the improvement of this process is not only the responsibility of the QA Team but of the whole project team, as it depends on development and project management support to implement all defined and proposed changes: "...to enable this change on the development process and generate meaningful results, QA Team should have the support from all project teams

and scopes..." - S3.

Specialist S4 described a vision focused on quality of deliverables and faster delivery: "...I see that the Team's main objective should be to improve and make the software delivery faster while maintaining high software quality deliverables..." - S4.

4.1.3 Categorized Gathered Knowledge

After interview summarization, authors categorized all gathered knowledge considering general objectives or specific improvements for the V1 of the development process:

QA Participation: Increase QA team's participation from early model development steps, with clearly defined activities requiring Project Management Team support since project initiation.

Feature or Requirements Analysis: A deeper software requirement analysis is needed, focused on how the dataset describes software issues and requirements, to verify and validate the dataset against software requirements and features.

Data Partitioning: Structured dataset sectioning into Training, Validation, and Testing partitions is crucial. The Testing partition should only be accessed by the QA Team, to test and validate the AI/ML model against defined test procedures.

Data Quality Review: An analysis process of the defined dataset to identify how to improve its significance and fit to software requirements and features, requiring data revisions and quality improvements.

Model Evaluation Quality Metrics: The team should identify how used metrics translate into quality metrics for the overall system, so that quality metrics can be correctly correlated and evaluated throughout the software development phase.

5 GATHERED KNOWLEDGE APPLIED ON MODEL DEVELOPMENT PROCESS

This research has the initial objective of providing a comprehensive guideline where we can define and understand QA professionals' activities in AI/ML enabled software development process. The guideline defines the steps, the actions, and how QA professionals can perform them throughout AI/ML software projects to support the final product and increase their quality aspects in the development process. The initial proposed version of the development process for AI/ML software was defined following the gathered knowledge from interviews and through specialists and expert reviews after the definition of the initial

development process. In addition, the initial proposed development process will take into account literature-defined best practices and issues, as well as their proposed solutions and fixes for some identified issues (Lwakatare et al., 2020; Serban et al., 2020). The initial version of the process is described in Figure 2, where for each development step the gathered knowledge was applied focusing on solving the needs for a better quality support process in the model development. The version focuses on defining how QA Team can develop activities under the same AI/ML model development process in order to enable software developers to produce AI/ML models with increased quality and focus on the software requirements and features.

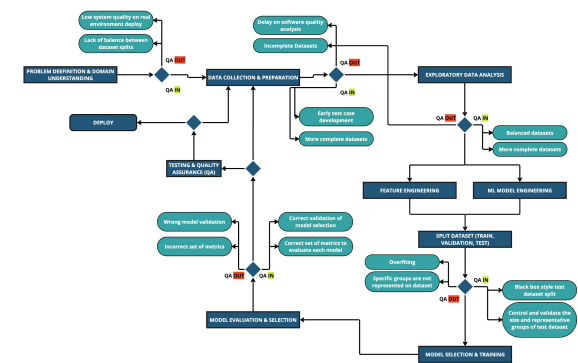


Figure 2: Generated Development process with the inclusion of gathered knowledge and requested changes from specialists.

The improved process covers the following phases with new quality steps: **Problem Definition & Domain Understanding, Data Collection & Preparation, Exploratory Data Analysis, Split Dataset (Train, Validation, Test), and Model Evaluation & Selection.**

5.1 Problem Definition & Domain Understanding

This is the phase where software development teams describe and understand the presented problem, its proposed solution, and the domain where it is allocated. Quality improvements focus on data analysis and data separation, ensuring all identified scopes of the problem domain are covered by relevant and significant data. QA members can also identify and develop the testing and evaluation process plan considering the real environment and problem domain, ensuring test procedures are correctly allocated.

For the *Problem Definition* phase, QA members should consider the following steps:

- **Environment Definition:** Evaluate the environment where the problem is located; understand real environment data that should be included in the dataset; support dataset categorization;
- **Data Capturing Specifications:** Analyze how environmental aspects influence the captured data;
- **Features and Requirements:** Support requirements elicitation and identify which data is needed to support defined requirements and features;

For the *Domain Understanding* phase, QA members should consider:

- **Data Type:** Identify the data types that need to be analyzed and collected for the AI/ML model training process;
- **Data Significance:** Analyze the significance of collected data in relation to the presented problem domain;

Regarding testing procedures, QA members should define the **Testing Environment**, **Testing Cases**, and **Test Case Applicability** based on the defined problem and domain, enabling more accurate and complete testing procedures. In this manner, QA Team will be able to correctly test the developed AI/ML model in terms of the defined problem and problem domain, considering important aspects of the software problem and where the solution will be applied. Moreover, QA Team will be able to focus on information that will render test procedures and datasets to be more complete and significant when covering all of the problem domain aspects.

5.2 Data Collection & Preparation

The QA Team should use knowledge from the problem and domain definition phase to support dataset collection and creation. Key activities include verifying:

- **Data Categories:** That the dataset covers all aspects of the problem definition;
- **Problem Coverage:** That entry types, data categories, and data types correctly cover the specified problem;
- **Dataset Size:** That the dataset contains enough data to produce relevant and sufficient splits;
- **Data Relevance:** That data is quality-compliant and problem-oriented;
- **Test Case Analysis:** Creating early test cases covering important software requirements from the collected data;

When QA Team correctly applies quality analysis in these early steps, the resulting software will have a more robust and complete AI/ML model, with a dataset more focused on the defined problem and its domain.

5.3 Exploratory Data Analysis

This phase focuses on understanding whether gathered and prepared data is relevant to the defined problem and suitable for training the AI/ML model. QA activities include:

- **Balance between Data Types:** Verify the sorting and balance of data for Training, Validation, and Testing;
- **Data Quality Aspects:** Validate that all data is ready for use with sufficient quality and size; advocate for new data when categories are incomplete or of insufficient quality;
- **Dataset Completeness:** Confirm that enough data was gathered or generated for each model development phase and data category identified;

When all the above steps are observed, the AI/ML model will more likely be precise and focused on the general problem, avoiding overfitting and other data-related issues.

5.4 Split Dataset

In this phase, the development team creates different dataset splits for training and validating the model. It is common to use an 80/20 split — 80% for training and 20% for validation (Goodfellow et al., 2016). The proposed change at this stage is for the QA team to create a third dataset split exclusive for testing, resulting in a **80/10/10 split** — 80% for training, 10% for validating, and 10% reserved for QA testing, partitioned from the validation split so as not to affect training data size. QA members should focus on:

- **Data Independency:** The Test dataset split must be completely independent from the development team;
- **Secure Access:** No one outside the QA Team should be authorized to access the testing split, preventing data leakage into training or model adjustments based on test data;
- **Dataset Balance:** All splits must contain the same proportion of data for each category;

5.5 Model Evaluation & Selection

In this phase, development teams evaluate and select the best trained model among different tested approaches. QA activities include:

- **Metrics Definition:** Define and identify which metrics are best fit for evaluating the model and translate them into relevant quality aspects for the software; defined metrics should be translatable into quality aspects of the model in terms of the applied problem solution;
- **Independent Evaluation:** Develop an independent evaluation process in terms of the evaluated model and metrics;

This phase depends heavily on model metrics evaluation and requires that QA members develop necessary knowledge on AI/ML metric evaluation for each type of model and domain.

6 SPECIALISTS DEVELOPMENT PROCESS REVIEW VALIDATION

In order to better understand and validate if the defined development process is useful, complete, and initially solves the identified problems, authors defined a Review process with the same specialists that identified the development process issues. This Review process is intended to validate the proposed development process and identify future gaps and improvements.

6.1 Review Validation Process

The review process involved the same four specialists who initially identified the issues with the traditional development process. Authors presented all the categorized data extracted from interviews and explained how each item of knowledge was applied to the initial proposed process, clarifying for each step the inputs, outputs, and respective QA activities. A section for questions was opened before moving to a structured review with a predefined list of questions. The evaluation followed the same rigor as the initial interviews.

6.2 Review Validation Results

Results are presented considering the structured questions developed for the structured interview and how these results can be categorized and analyzed for the improvement of the proposed process. The full list of

questions and extra information about the interviewed subjects are available in the supplementary technical report for this paper (Manzoni et al., 2026).

6.2.1 Question 1

The general opinion was that Problem Definition and Domain Understanding are the most important early steps for QA contribution: *"I understand that the initial development steps of the problem definition and domain understanding section ... are the ones that provide the whole context of the software being developed."* - S1. Specialists S2, S3, and S4 agreed with this vision. However, S2 and S3 noted that QA allocations are very late on project planning: *"... probably with this definition we could face and solve the limitations of early allocation of QA members ... as now QA activities will be clear and stated since the project start."* - S3.

6.2.2 Question 2

General responses were that Problem Definition and Domain Understanding phases commonly raise future problems related to insufficient or irrelevant data, making AI/ML models unfit for the defined problem: *"... this is not the most time consuming task, but it surely is the most critical one ... any misjudgment in this phase will reflect on the rest of the project ..."* - S1. S2 reinforced that errors at this phase lead to models being unusable: *"... any errors here will make models unusable for the defined problem and reworks would be needed ... we are talking about weeks/sprints of rework which is very critical ..."* - S2.

6.2.3 Question 3

General answers were that the development team could provide input data and support QA members through data elicitation and automatic generation for data augmentation: *"Development team can provide these processes and help QA teams on this focus of the datasets and inputs for the long run"* - S4. S2 and S3 also identified that the development team itself lacks defined data augmentation processes: *"... we need to define these processes inside the development team as well as we currently do not have them ..."* - S2 and *"I believe that the Development Team has the knowledge and abilities to supply these tools for the QA Team even though they are not existent for this task now!"* - S3.

6.2.4 Question 4

Opinions were mixed. The majority evaluated early test cases positively, noting they may uncover issues

on problem definition earlier: *"They will surely increase our confidence in the elicited dataset and the validation process of the dataset before training the model!"* - S1 and *"I see a great potential in the development of early test cases to guide development teams and the project quality dissemination ... I see that they can be a real weapon to develop the ownership sense in all project members."* - S3. However, S4 raised the concern that early test cases may cause rework as the problem definition can change: *"Even though this development may significantly support the development process, I fear that this step will produce disposable test cases that will be redeveloped as new information about the system is gathered"* - S4.

6.2.5 Question 5

Specialists evaluated that QA members should be capable of independently evaluating data and model metrics: *"... when considering these metrics, I understand that QA members will be able to handle them independently from the model development team, so that their quality evaluation is isolated and will not be biased by development model evaluations!"* - S1 and *"... I expect that a QA can actually tell me how these metrics are being evaluated in terms of project quality and that they can be independent from me or the development team ..."* - S2.

6.2.6 Question 6 and 7

The AI/ML development specialist indicated that adjusting the validation split from 20% to 10%/10% should not negatively affect training, though domain-specific models may have restricted datasets requiring caution. Specialists agreed that the testing split must never be accessible by the development team; however, QA must provide general data characteristics so that the development team can focus fixes on specific code sections, with this description available since the beginning of testing procedures.

6.2.7 Question 8 and 9

Specialists identified the following as minimum expected knowledge for QA members when evaluating AI/ML models: *Accuracy, Precision, Recall, F1-Score, AUC-ROC, MAE, MSE, and RMSE*. Each metric is selected depending on whether the task involves classification or regression.

6.2.8 Question 10

S2 stated that *"...even though we have not applied the proposed model on a project, the general perception is that this process might fix some important issues*

and gaps that development teams encounter daily on AI and ML model development phases...". S1 noted the need for more specific tasks and standardized procedures. S4 emphasized the process helps justify the institute's readiness for higher quality delivery: *"Looking into the process considering the company view on delivering higher quality systems, this process will surely help to justify that our institute is ready to deliver higher quality systems and that all sections of the development of the model is being covered!"* - S4. S3 raised a crucial concern regarding knowledge gaps: *"When looking into this initial process ... I think that QA members will have a hard time to apply the steps as it is, and members should be more knowledgeable on AI/ML model development processes to be able to contribute on it ..."* - S3.

Overall, the review validation confirms that the proposed process addresses the primary gaps identified by specialists, particularly regarding early QA involvement, dataset governance, and independent model evaluation. The feedback also highlights the need for clearer task operationalization and QA skill development, which directly informs the next iteration of the process.

7 THREATS TO VALIDITY

The threats to validity of this study are discussed according to the four categories defined by Wohlin (Wohlin et al., 2024).

Internal Validity: Considering internal validity, there could be a conflict of interest on specialists' opinions and their latter evaluation of the proposed development process as they reviewed a process developed based on their own inputs and opinions. This threat is partially mitigated by the specialists' direct stake in improving the development process.

External Validity: The proposed process is not intended to generalize to all AI/ML models or domains. However, it was the objective of this initial version to focus on specific models and domains inside the company enabling the process to be expanded in later versions.

Conclusion Validity: Considering the view of only four specialists they may not represent the opinion from all the population of AI/ML developers and researchers. In an attempt to minimize this threat, each specialist had different areas of activities and expertise. Also, specialists had expertise in a range of AI projects, spanning different contexts of application. Moreover, this paper does not intend to define a final version of the development process, but to present the explored initial motives and industry chal-

lenges that will drive the research to find a conclusion.

Construct Validity: For this type of threat, the initial base AI development process model considered have been used by the company for the last decade that also was previously developed based on literary knowledge.

8 CONCLUSIONS AND FUTURE WORK

We have presented the initially proposed AI/ML Model Development Process, considering inputs and feature changes gathered from specialists and their experiences with traditional processes. The proposed process addresses five main categories of issues identified from interviews: (i) Increase QA Participation on Development Process; (ii) Introduce Model Feature or Requirements Analysis by QA; (iii) Data Partitioning for Testing; (iv) Data Quality Reviews and Checks; and (v) Model Evaluation Quality Metrics and Aspects.

After applying improvements considering these five categories, a process with specific QA activities for each development phase was proposed and subsequently evaluated by the same four specialists. From this evaluation, improvements were raised: more specified processes and activities for the QA Team (S1, S2, S3) and clearer communication of process benefits and outcomes (S4).

This work contributed to bringing to light the discussions about how AI/ML model development processes are faced in industry and the challenges faced by the professionals. Furthermore, this work proposes and envisions an initial improved development process that considers the pain points from development and management teams in industry.

Future work includes evaluating the process across multiple and varied companies to gather more general information; performing a more rigorous evaluation with different real-world projects to verify whether improvements were effective; and conducting a longitudinal validation study to understand the real-world effects of the applied process in software development projects.

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